# Green Cloud Scheduler

## Energy-Aware Workload Consolidation for IaaS Control Planes

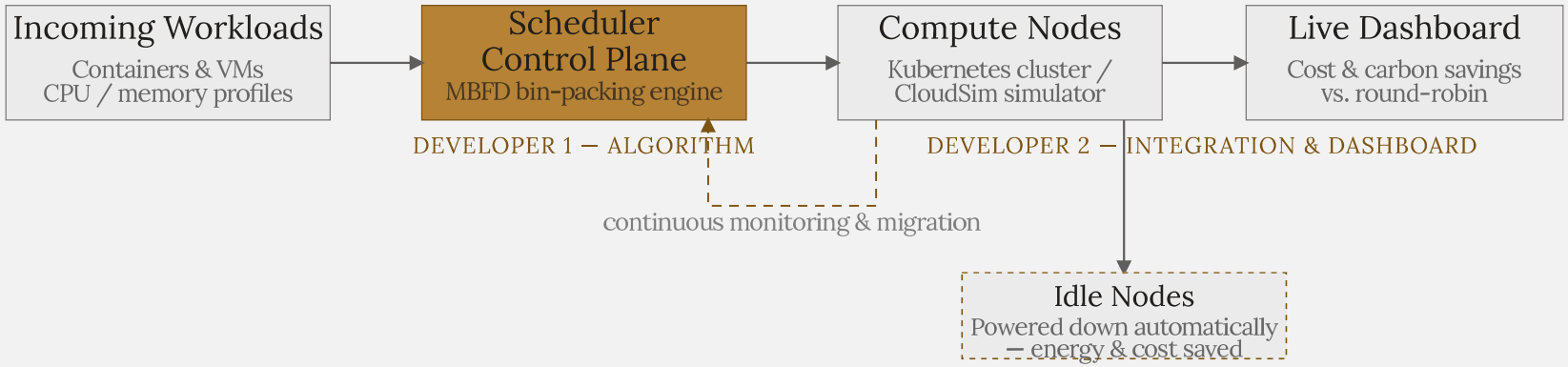
Base research paper: Reddy et al., "ROA: Energy and Environment Aware Task Scheduling in Cloud Computing," IEEE ICSCDS, 2025

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# Problem Statement

- A server running at 10% capacity draws nearly as much power as one running at 100% — idle headroom is not free.
- FinOps and Green Computing now dominate cloud strategy: consolidate workloads tightly, then shut down unused hardware.
- Standard round-robin schedulers spread load evenly across every node, keeping the whole fleet awake for no reason.
- Goal: continuously repack active workloads onto the fewest physical nodes possible, and power down what's left over.

# System Architecture




The scheduler continuously monitors load, migrates active workloads via MBFD bin-packing, and powers down any node left empty — reporting the resulting cost and carbon savings against a round-robin baseline.

# Novelty — vs. Base Paper

- The base paper optimizes a one-shot task-to-VM mapping with a tunable metaheuristic (GA/PSO/ACO-style search); we run a lightweight bin-packing heuristic (MBFD) as a continuous, always-on control loop — no population tuning, no offline optimization run per batch.
- Their algorithm ends at "assign the task"; ours goes one step further and actually powers down or scales down every physical node left empty after consolidation — closing the loop from decision to real energy action.
- They validate purely in CloudSim against a static HPC trace (CEA-Curie); we target CloudSim and a live Kubernetes cluster, so the same scheduler can run on real infrastructure, not just simulated jobs.

# Novelty — Benchmark & Extension

- They benchmark against other metaheuristics (GA, PSO, ACO); we benchmark against round-robin — the scheduler most real clusters actually run today — making our savings story directly relevant to a practitioner.
- We add a free, zero-infrastructure-cost extension beyond the paper's scope: carbon-intensity-aware scheduling, using a free public grid-carbon API to time-shift delay-tolerant batch jobs to greener hours on top of our consolidation engine.